

Renal Reference — Intrinsic AKI by Compartment

Intrinsic AKI by Compartment — Fast Mapping

Injured compartment	Think of	Classic triggers	Urine / lab clue	Board meaning
Tubules	ATN — ischemic or nephrotoxic	Hypotension/shock, sepsis, contrast, aminoglycosides, rhabdo, toxins	Muddy brown granular casts ± renal tubular epithelial cells/casts	Tubules are injured and cannot reabsorb Na ⁺ well → FeNa often >1%, but not always
Interstitium	AIN	New drug: PPI, NSAID, antibiotics, immune checkpoint inhibitors; infections; autoimmune disease	WBC casts, sterile pyuria ± eosinophils	Interstitial inflammation
Glomeruli	GN / vasculitis	Autoimmune disease, recent infection, ANCA disease, anti-GBM, lupus, low-complement disease	RBC casts - dysmorphic RBCs ± proteinuria	True glomerular hematuria
Renal vessels	TMA / atheroemboli / malignant HTN / large-vessel occlusion	TTP/HUS, malignant HTN, catheterization/aortic manipulation, APS, cocaine, anti-VEGF, calcineurin inhibitors	MAHA + thrombocytopenia, schistocytes, ↑LDH; livedo/blue toes; severe HTN	Vascular injury or occlusion causing AKI
Tubules — pigment injury	Pigment nephropathy	Rhabdomyolysis, hemolysis	Heme-positive dipstick with few/no RBCs	Myoglobin or hemoglobin is detected as “heme”

1. Tubules row

WHAT YOU SEE

Row 1 compartment 'Tubules'

WHAT IT ENCODES

The tubular compartment = acute tubular necrosis (ATN), the single most common cause of hospital-acquired intrinsic AKI. ATN has two flavors: ISCHEMIC (prolonged prerenal hypoperfusion that crosses into tubular cell death) and NEPHROTOXIC (direct toxins). It evolves through initiation → maintenance (oliguric, days–weeks) → recovery (polyuric) phases.

BOARD TRAP

Prerenal AKI and ATN are unrelated diagnoses. WRONG — they are a continuum; untreated prerenal hypoperfusion progresses to ischemic ATN once tubular cells die. The discriminator is reversibility with fluids (prerenal) vs not (ATN) plus muddy-brown casts and FENa >2% in ATN.

2. Tubules triggers

WHAT YOU SEE

Tubules triggers: hypotension/shock, sepsis, contrast, aminoglycosides, rhabdo, toxins

WHAT IT ENCODES

Ischemic triggers: hypotension/shock, sepsis, major surgery, hemorrhage. Nephrotoxic triggers: contrast, aminoglycosides (5–10 day lag, nonoliguric, with Mg/K wasting), amphotericin B, cisplatin, vancomycin, and endogenous pigments (myoglobin in rhabdo, hemoglobin in hemolysis). Tumor lysis (uric acid) and ethylene glycol (oxalate crystals) also injure tubules.

BOARD TRAP

Vancomycin and aminoglycoside nephrotoxicity look identical. WRONG — aminoglycoside ATN is classically NONoliguric with proximal tubular wasting (hypoMg, hypoK); pinning the mechanism requires the exposure timeline and urine sediment, not assuming all 'antibiotic AKI' is the same.

3. Tubules urine clue

WHAT YOU SEE

Tubules urine: muddy brown granular casts ± renal tubular epithelial cells/casts

WHAT IT ENCODES

Hallmark sediment: MUDDY-BROWN granular casts and renal tubular epithelial cell casts. Indices favor ATN: FENa >2%, urine Na >40, urine osm ~300 (isosthenuria — injured tubules can't concentrate), BUN:Cr ~10–15 (normal). Note the important exceptions where ATN shows FENa <1%: early sepsis, contrast, and rhabdo/pigment.

BOARD TRAP

FENa is always >2% in ATN. WRONG — early sepsis, contrast-induced AKI, and pigment nephropathy are intrinsic/tubular but keep FENa <1%; muddy-brown casts and clinical context, not FENa alone, clinch ATN.

4. Interstitium row

WHAT YOU SEE

Row 2 compartment 'Interstitium' → AIN

WHAT IT ENCODES

The interstitial compartment = acute interstitial nephritis (AIN), an immune/hypersensitivity inflammation of the renal interstitium. Classic (but present in <10–30%) triad: fever, rash, eosinophilia. Onset is typically days to weeks after a new drug; AKI is often accompanied by sterile pyuria and mild proteinuria.

BOARD TRAP

Absence of the fever-rash-eosinophilia triad rules out AIN. WRONG — the full triad appears in a minority of cases; AIN remains the diagnosis with a compatible drug exposure, WBC casts/sterile pyuria, and AKI. Renal biopsy is definitive when needed.

5. Interstitium triggers

WHAT YOU SEE

AIN triggers: new drug — PPI, NSAID, antibiotics, immune checkpoint inhibitors; infections; autoimmune

WHAT IT ENCODES

Drug-induced (most common): PPIs, NSAIDs, beta-lactams (penicillins, cephalosporins), sulfonamides, rifampin, allopurinol, and immune-checkpoint inhibitors (a growing oncology cause). Also infections (Legionella, Leptospira) and autoimmune (sarcoidosis, Sjögren, lupus, TINU). NSAID-AIN is unique in causing AIN PLUS nephrotic-range proteinuria (minimal change).

BOARD TRAP

Stopping the offending drug and giving steroids isn't urgent in AIN. WRONG — the key intervention is prompt withdrawal of the culprit drug; early corticosteroids may aid recovery, and delay risks irreversible interstitial fibrosis.

6. Interstitium urine clue

WHAT YOU SEE

AIN urine: WBC casts, sterile pyuria ± eosinophils

WHAT IT ENCODES

Sediment: WBC casts and sterile pyuria (pyuria with negative culture) ± hematuria and mild proteinuria. Urine eosinophils (Hansel stain) are the classic teaching clue but are neither sensitive nor specific. AKI with a bland-to-pyuric sediment after a new PPI/NSAID/antibiotic should raise AIN.

BOARD TRAP

Negative urine eosinophils excludes AIN, and positive eosinophiluria confirms it. WRONG on both counts — eosinophiluria has poor sensitivity AND specificity (also seen in atheroembolic disease, UTI, GN); diagnosis rests on drug history, sterile pyuria/WBC casts, and biopsy when needed.

7. Glomeruli row

WHAT YOU SEE

Row 3 compartment 'Glomeruli' → GN/vasculitis

WHAT IT ENCODES

The glomerular compartment = glomerulonephritis / vasculitis, a NEPHRITIC process: inflammation of the glomerular tuft causing hematuria with RBC casts, dysmorphic RBCs, hypertension, edema, and sub-nephrotic proteinuria. Rapidly progressive GN (RPGN) with crescents is the can't-miss emergency requiring urgent biopsy and immunosuppression.

BOARD TRAP

Glomerular AKI is identified by FENa/urine Na like ATN. WRONG — GN often shows a prerenal-pattern FENa <1% despite intrinsic injury; the discriminator is the ACTIVE sediment (RBC casts, dysmorphic RBCs) plus serologies, not the sodium indices.

8. Glomeruli triggers

WHAT YOU SEE

GN triggers: autoimmune disease, recent strep infection, ANCA disease, anti-GBM, lupus, low-complement

WHAT IT ENCODES

Serologic workup localizes the GN: ANCA (granulomatosis with polyangiitis, microscopic polyangiitis — pauci-immune); anti-GBM (Goodpasture — linear IgG, hemoptysis + AKI); ANA/dsDNA + LOW complement (lupus nephritis); low complement also in post-strep GN (recent pharyngitis/impetigo, weeks prior) and MPGN/cryoglobulinemia (hepatitis C). Normal complement: IgA nephropathy, ANCA, anti-GBM.

BOARD TRAP

A young adult with hematuria days after a sore throat has post-streptococcal GN. WRONG — synpharyngitic hematuria (within 1–5 days of URI) is IgA nephropathy; post-strep GN has a 1–3 WEEK latency with LOW complement that normalizes by ~8 weeks. Complement levels and timing separate them.

9. Glomeruli urine clue

WHAT YOU SEE

GN urine: RBC casts + dysmorphic RBCs ± proteinuria

WHAT IT ENCODES

Hallmark: RBC casts and DYSMORPHIC (acanthocyte) RBCs = true glomerular bleeding (nephritic). Proteinuria is usually sub-nephrotic (<3.5 g/day); nephrotic-range with bland sediment instead points to a non-inflammatory glomerulopathy (membranous, FSGS, minimal change). RBC casts are essentially pathognomonic for GN/vasculitis.

BOARD TRAP

Hematuria with RBC casts could be from a kidney stone or bladder cancer. WRONG — RBC casts form only in the tubules and signal GLOMERULAR bleeding; isomorphic RBCs without casts suggest a urologic (stone/tumor) source. The presence of casts redirects the workup to GN, not urology.

10. Renal vessels row

WHAT YOU SEE

Row 4 compartment 'Renal vessels' → TMA/atheroemboli/malignant HTN/large-vessel occlusion

WHAT IT ENCODES

The vascular compartment splits into MICROvascular (thrombotic microangiopathy — TTP, HUS, DIC, scleroderma renal crisis, malignant HTN, APS) and MACROvascular (atheroembolic/cholesterol emboli, renal artery thrombosis/dissection, large-vessel occlusion). Each has distinct systemic clues beyond the kidney.

BOARD TRAP

Microangiopathic anemia with low platelets and AKI is automatically TTP needing plasma exchange. WRONG — distinguish TTP (ADAMTS13 <10%, neuro signs) from HUS (Shiga-toxin diarrhea/E. coli O157, children — supportive care, NOT routine plasma exchange) and atypical HUS (complement → eculizumab). Treatment diverges sharply.

11. Renal vessels clue

WHAT YOU SEE

Vessel urine/labs: MAHA + thrombocytopenia, schistocytes, livedo/blue toes, severe HTN

WHAT IT ENCODES

TMA clues: microangiopathic hemolytic anemia (schistocytes), thrombocytopenia, elevated LDH, low haptoglobin, normal coags (vs DIC). Atheroembolic clues: livedo reticularis, blue/purple toes with intact pulses, EOSINOPHILIA + low complement + eosinophiluria, days–weeks AFTER a vascular procedure/angiography or anticoagulation. Malignant HTN: BP often >180/120 with retinal/CNS findings.

BOARD TRAP

AKI plus eosinophilia after a cardiac catheterization is contrast nephropathy or AIN. WRONG — cholesterol (atheroembolic) AKI presents days-to-weeks post-procedure with blue toes, livedo, transient eosinophilia, and low complement; contrast AKI peaks at 3–5 days and resolves, and AIN is drug-triggered. The skin findings and timing distinguish them.

12. Pigment injury row

WHAT YOU SEE

Bottom row 'Tubules — pigment injury' → pigment nephropathy

WHAT IT ENCODES

Pigment nephropathy is a TUBULAR subtype: rhabdomyolysis releases myoglobin (or massive hemolysis releases hemoglobin) which is directly tubulotoxic, forms casts with Tamm-Horsfall protein, and causes intrarenal vasoconstriction. Clues to rhabdo: CK in the thousands–hundreds of thousands, hyperkalemia, hyperphosphatemia, early HYPOcalcemia then late hypercalcemia, and a LOW/normal BUN:Cr (muscle dumps creatinine).

BOARD TRAP

Pigment (rhabdo) AKI shows a high BUN:Cr and high FENa like typical ATN. WRONG — rhabdo gives a LOW/normal BUN:Cr (direct creatinine release) and can show FENa <1% (intact tubular avidity early); the giveaways are CK elevation and heme-positive dipstick without RBCs.

13. Pigment urine clue

WHAT YOU SEE

Pigment urine: heme-positive dipstick with few/no RBCs

WHAT IT ENCODES

Signature finding: a dipstick POSITIVE for blood/heme but FEW or NO RBCs on microscopy — because the dipstick detects the heme moiety of myoglobin and hemoglobin, not intact erythrocytes. Pair with red-brown ('tea/cola-colored') urine, markedly elevated CK (rhabdo) or low haptoglobin/high LDH (hemolysis) to confirm.

BOARD TRAP

A heme-positive dipstick always means true hematuria. WRONG — myoglobin and free hemoglobin also turn the dipstick positive; if microscopy shows no RBCs, think pigment nephropathy and send CK, haptoglobin, and LDH rather than chasing a urologic bleeding source.

14. Board meaning column

WHAT YOU SEE

Right column 'Board meaning'

WHAT IT ENCODES

The exam logic: each compartment maps to a SIGNATURE sediment, so the urine microscopy IS the localizing test for intrinsic AKI — muddy-brown casts = ATN (tubules); WBC casts/sterile pyuria = AIN (interstitium); RBC casts/dysmorphic RBCs = GN (glomeruli); schistocytes/blue toes = vascular; heme-positive without RBCs = pigment. Order: assess pre/post-renal first, then use sediment to pin the intrinsic compartment.

BOARD TRAP

You can localize intrinsic AKI from FENa and BUN:Cr alone. WRONG — those indices mainly separate prerenal vs ATN and have many exceptions; the URINE SEDIMENT is what distinguishes ATN vs AIN vs GN vs vascular vs pigment. Always examine the sediment in intrinsic AKI.